

```

apiVersion: v1
kind: ConfigMap
metadata:
  name: holiday-list
data:
  2023-01-01: New Year's Day
  2023-01-02: New Year's Day observed
  2023-01-16: Martin Luther King Jr. Day
  2023-02-20: Presidents' Day
  2023-05-29: Memorial Day
  2023-07-04: Independence Day
  2023-09-04: Labor Day
  2023-10-09: Columbus Day
  2023-11-10: Veterans Day observed
  2023-11-23: Thanksgiving Day
  2023-12-25: Christmas Day

```

Figure 1: ConfigMap with list of holidays

```

#!/bin/bash

set -e

holiday_list=$(cat /etc/holidays.txt)

if grep -q "$(date +%Y-%m-%d)" <<< "$holiday_list"; then
  echo "$(date +%Y-%m-%d) is a holiday, exiting"
  exit 1
else
  echo "$(date +%Y-%m-%d) is not a holiday, continuing"
fi

```

Figure 2: Script to check current date

```

apiVersion: batch/v1beta1
kind: CronJob
metadata:
  name: my-cronjob
spec:
  schedule: "0 0 * * *"
  jobTemplate:
    spec:
      template:
        spec:
          containers:
            - name: my-container
              image: my-image
              command: ["/bin/bash", "-c", "/bin/holiday-
volumeMounts:
            - name: holiday-list
              mountPath: /etc/holidays.txt
              subPath: holidays.txt
              restartPolicy: OnFailure
          volumes:
            - name: holiday-list
              configMap:
                name: holiday-list

```

Figure 3: Mounting the ConfigMap

the actual command `/bin/my-command`. The Bash script reads the holiday list from the mounted ConfigMap file `/etc/holidays.txt` and exits with an error code if the current date is a holiday.

Sample code for scheduling jobs

- a. To run a job every day at a specific time you can use the sample code given below:

```

apiVersion: batch/v1beta1
kind: CronJob
metadata:
  name: my-cronjob
spec:
  schedule: "*/5 * * * *"
  jobTemplate:
    spec:
      template:
        spec:
          containers:
            - name: my-container
              image: my-image
              command: ["/bin/sh", "-c", "echo hello"]
              restartPolicy: OnFailure
          Metadata:
            labels:
              app: my-app
            successfulJobsHistoryLimit: 3
            failedJobsHistoryLimit: 3

```

In the above code, a CronJob is scheduled to run every 5 minutes (line 6). The job runs a simple command to echo 'hello'. The restartPolicy is set to OnFailure, which means that the job will be restarted if it fails.

- b. To run a job once a week, use the following format in the above code, line 6:

```
0 5 * * 0
```

The job will run at 5:00 a.m. every Sunday. The last digit (0) indicates the day of the week, where Sunday is 0 and Saturday is 6.

- c. To run a job at the end of every month, use the following format in the above code, line 6.

```
0 0 28-31 * *
```

The job will run at midnight on the 28th, 29th, 30th, or 31st day of the month.

- d. To prevent a job from running on holidays, you'll need to create a list of holidays and exclude those days from the Cron schedule. Here's an example: